FYP MEETING LOG

Supervisor Meeting 3 of 3

Agent-Based Game Analysis in Unity Based Upon Curiosity and RND Exploration Using PPO and SAC
Student: Raza Ullah Khan (NUM-BSCS23-28) | Supervisor: Dr. Malik M Ali Shahid | Co-Supervisor: Mr. Abdul Rafay | Industrial Partner: Inspired Square

Meeting No.	06 (Supervisor Meeting 3)
Meeting With	Supervisor - Dr. Malik M Ali Shahid
Date	Tuesday, 3 November 2026
Time	2:30 PM - 3:00 PM
Duration	30 minutes
Mode / Venue	Supervisor's office, Namal University Mianwali
Attendees	Raza Ullah Khan (Student); Dr. Malik M Ali Shahid (Supervisor)
Milestone	FN-02 (Month 1, Weeks 3-4): Hummingbirds 3D Flight Area & Flower Setup

1. Purpose of the Meeting

A review of the first month: what is done, what problems we see coming, and the plan for the next two weeks.

2. Agenda

1. Progress in the first month.
2. The ML-Agents version issue raised by the co-supervisor meeting.
3. Main risks and how we handle them.
4. README and the state of the GitHub repository.
5. Early thoughts on the second game.

3. What We Discussed

- **Progress.** Unity, Python and the trainer are installed. The repository is on GitHub with the README, proposal documents, Unity package list, project settings, .gitignore and requirements file. The metric definitions are written and the proposal references are corrected. Paper summaries are almost done.
- **Version issue.** I explained that RND is not built into mlagents 0.16.1. Sir agreed with the plan: keep the current setup for baselines and decide before the exploration stage. He added that the decision and its reason must be written in the report's methodology section.
- **Computing time.** Six conditions with five seeds means 30 runs in the main game, plus the selected runs in the second game. Sir asked me to time one baseline run first and then make a realistic training schedule.
- **README.** The README should tell a new reader in one page what the project is, the six conditions, the metrics, the tools and the team. Sir checked the README on GitHub and found it clear.
- **Second game.** Not decided yet. A simple platformer-style game with a sparse reward could be a good contrast to Hummingbirds, because exploration should matter more there. The final choice will be made around Month 5.

4. Small Implementation Part

We reviewed the README structure and wrote down the main risks with a simple plan for each.

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README.md  (current structure on GitHub)
  1. Overview            - what the project studies, in plain words
  2. Experiment matrix   - PPO / SAC  x  Baseline / Curiosity / RND
  3. Evaluation metrics  - extrinsic return (main), progress, coverage
  4. Deliverables        - trained agents, code + YAML, W&B, report
  5. Tools               - Unity, C#, ML-Agents, Python, W&B, Git
  6. Timeline            - 8 months, current stage: environment setup
  7. Team                - student, supervisor, co-supervisor, partner
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Risk	Effect	What we will do
Old ML-Agents version has no RND	Cannot run Conditions 3 and 6	Decide upgrade or own RND before FN-06
30+ runs take too long	Schedule slips	Time one run first; train headless with parallel envs; run overnight
Agent learns a shortcut	Wrong conclusions	Watch videos of trained agents, not only graphs
Different settings by mistake	Unfair comparison	One shared YAML; only reward_signals changes
Second game not ready	No transfer check	Pick a simple game early (by Month 5)

5. Decisions Taken

- Milestones FN-01 and FN-02 (setup and first month) are reviewed and accepted.
- Work on main and develop goes only through Pull Requests from now on.
- The version decision will be recorded with its reason in the methodology chapter.
- Start FN-03 (agent observations, continuous actions and keyboard testing) now.

6. Action Items

#	Action Item	Responsible	Due Date
1	Finish the paper summaries and share them with the supervisor.	Raza	13 Nov 2026
2	Time one short training run on the lab PC.	Raza	After FN-04
3	Set up the agent's observations and actions and test it with the keyboard (FN-03).	Raza	24 Nov 2026
4	Share the ML-Agents upgrade comparison with the supervisor.	Raza	18 Nov 2026

7. Other Notes

- Sir asked to keep all meeting logs together in the docs folder.

8. Next Meeting

Tuesday, 10 November 2026 - progress on FN-03 and the ML-Agents version decision.